Employee Attrition Prediction – Executive Overview

HR Analytics Insights

This study examined workforce data to uncover the drivers behind employee turnover and provide actionable guidance for Human Resources. By analyzing historical records, we identified patterns distinguishing employees who remained with the company from those who chose to leave.

The organization's overall attrition rate stands at **16.12%**, meaning roughly one in six employees exits. Turnover is notably higher within the **Sales Department**, with **Sales Representatives** and **Laboratory Technicians** showing the greatest vulnerability. A clear link was observed between poor **Work-Life Balance** ratings and higher attrition, underscoring the importance of employee well-being in retention.

Predictive modeling highlighted several influential factors: **Monthly Income**, **Age**, **Total Working Years**, **Tenure at the Company**, **Commute Distance**, **Overtime frequency**, and **Job Role**. These findings confirm that attrition is shaped by a combination of career progression, workload, and personal circumstances—not salary alone.

To address these risks, HR should focus retention strategies on departments and roles with elevated turnover, especially employees working excessive overtime or those in the early stages of tenure. Initiatives such as regular engagement surveys, career growth programs, flexible work policies, and targeted retention discussions can help reduce voluntary exits.

It is important to note that while the predictive model offers valuable guidance, it should complement—not replace—human judgment. Factors like organizational culture, leadership quality, and external market conditions, which were not captured in the dataset, also play a role in employee decisions.

In summary, this project demonstrates how data-driven insights can empower HR teams to identify at-risk employees, take proactive measures, and ultimately lower the financial and operational impact of attrition.